Effective Nuclear Charge and Shielding: Worksheet key

1. How many core electrons does nitrogen (Z = 7) have? 2

2. Estimate Z_eff for magnesium (Z = 12). 2

3. Estimate Z_eff for chlorine (Z = 17). 7

4. How many core electrons does potassium (Z = 19) have? 18

5. Which ring number holds the outer electron of boron (Z = 5)? 2

6. Estimate Z_eff for sulfur (Z = 16). 6

7. lithium (Z_eff 1, n 2, pull 1/4) vs potassium (Z_eff 1, n 4, pull 1/16): lithium holds harder.

8. sodium (Z_eff 1, n 3, pull 1/9) vs potassium (Z_eff 1, n 4, pull 1/16): sodium holds harder.

9. magnesium (Z_eff 2, n 3, pull 2/9) vs chlorine (Z_eff 7, n 3, pull 7/9): chlorine holds harder.

10. sodium (Z_eff 1, n 3, pull 1/9) vs sulfur (Z_eff 6, n 3, pull 6/9): sulfur holds harder.

11. Sodium: +11, rings 2, 8, 1; core rings (2 and 8) shaded; Z_eff = 1.

12. Na 1, Mg 2, Al 3, Si 4, P 5, S 6, Cl 7, Ar 8

13. Both have Z_eff = 1, but the outer electron of K is in ring 4 and Na in ring 3. The farther electron is pulled less, so potassium's comes off more easily. More protons are cancelled by more core electrons.

14. The core stays the same and Z_eff rises by one each step, so the outer electrons are pulled in harder: the atoms should get smaller across the period (same ring number, bigger pull).

Total: 30 points.